

Research Article

Enhanced in vitro type 1 conventional dendritic cell generation via the recruitment of hematopoietic stem cells and early progenitors by Kit ligand

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In vitro culture of bone marrow (BM) with Fms-like tyrosine kinase 3 ligand (Flt3L) is widely used to study development and function of type 1 conventional dendritic cells (cDC1). Hematopoietic stem cells (HSCs) and many progenitor populations that possess cDC1 potential in vivo do not express Flt3 and thus may not contribute to Flt3L-mediated cDC1 production in vitro. Here, we present a KitL/Flt3L protocol that recruits such HSCs and progenitors into the production of cDC1. Kit ligand (KitL) is used to expand HSCs and early progenitors lacking Flt3 expression into later stage where Flt3 is expressed. Following this initial KitL phase, a second Flt3L phase is used to support the final production of DCs. With this two-stage culture, we achieved approximately tenfold increased production of both cDC1 and cDC2 compared to Flt3L culture. cDC1 derived from this culture are similar to in vivo cDC1 in their dependence on IRF8, ability to produce IL-12, and induction of tumor regression in cDC1-deficient tumor-bearing mice. This KitL/Flt3L system for cDC1 production will be useful in further analysis of cDC1 that rely on in vitro generation from BM.

Keywords: Dendritic cell · Flt3L · KitL · HSC · Tumor immunology



Additional supporting information may be found online in the Supporting Information section at the end of the article.

Introduction

Dendritic cells (DCs) comprise several related hematopoietic lineages that function in innate and adaptive immune responses

[1]. Conventional DCs (cDCs) comprise two major subsets that serve as antigen-presenting cells to prime naïve T cells and activate innate lymphoid cells [2–4]. cDC1 require the transcription factors NFIL3, ID2, BATF3, and IRF8 for development and specialize in cross-presentation to prime CD8 T cells against many intracellular pathogens and tumors [3]. cDC2 supports both Th17 [5, 6] and Th2 responses to various other pathogens [7–9]. A third DC subset, plasmacytoid DC (pDC), serves as sentinels of

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viral infection and augment T-cell responses by the production of cytokines [10].

DCs can be generated from bone marrow (BM) progenitors cultured with the activating ligand for Fms-like tyrosine kinase 3 (Flt3) [11]. Flt3 encodes a class III receptor tyrosine kinase closely related to Kit and macrophage colony-stimulating factor 1 receptor (M-CSFR or Fms) [12]. Identification of Flt3 ligand (Flt3L) [13] inspired attempts using it in expanding hematopoietic cells. Subcutaneous Flt3L administration in mice dramatically expanded CD11c⁺ MHCII⁺ DC populations [14]. Flt3L alone could generate CD11c⁺ DCs from densely plated murine BM in vitro [15] and generates cDC1, cDC2, and pDC [11, 16]. cDC1 produced by this method resemble in vivo cDC1 in several ways, including the developmental requirement for IRF8 [11] and BATF3 [17], and induction of IL-12 by *Toxoplasma gondii* soluble tachyzoite antigen (STAg) [18]. However, Flt3L cultures of BM produce relatively small numbers of cDC1 compared with the substantially greater numbers of monocyte-derived DCs (MoDCs) generated GM-CSF culture [19, 20]. As a result, investigating cDC1 on a large scale has been challenging.

Hematopoietic stem cells (HSCs) were originally described as Lineage⁻ Sca1⁺ Kit⁺ (LSK) cells [21, 22] which now include self-renewing long-term HSC (LT-HSC), short-term HSC (ST-HSC), and nonself-renewing multipotent progenitors (MPPs) [23–26]. MPPs can be further divided into MPP2, MPP3, and MPP4 [25, 26]. Among these, LT-HSC, ST-HSC, MPP2, and MPP3 do not express Flt3 [26]. As these populations all have in vivo DC potentials [27] and all express Kit, we wondered whether Kit ligand (KitL or SCF) could enhance in vitro cDC1 production via the recruitment of these HSCs and MPPs into Flt3L culture.

Results

KitL treatment leads to enhanced in vitro production of all DC subsets

We used two conditions to test whether KitL can enhance in vitro cDC1 generation. In a sequential protocol, we cultured BM in KitL for 3 days, followed by treatment with Flt3L for another 8 days. In a combined protocol, we cultured BM with KitL plus Flt3L for 3 days, followed by treatment with Flt3L for another 8 days (Fig. 1A). Both KitL/Flt3L protocols generated significantly more live cells and cDC1 compared to our standard 8-day culture with Flt3L alone (Fig. 1B and Supporting Information Fig. 1A–E). We also tested the effect of the initial plating density of BM on the overall output of cDC1. Using high density BM plating standard for our Flt3L protocol (2.5×10^6 cells/mL), KitL showed little impact on enhancing cDC1 output (Supporting Information Fig. 1A and B). By contrast, using low density BM plating (6.25×10^5 cells/mL), KitL caused a substantial increase in cDC1 output for both sequential and combined KitL/Flt3L protocols (Supporting Information Fig. 1A and B). Time course analyses showed that both KitL/Flt3L conditions increased the production of cDC1,

cDC2, and pDC at all time points compared with Flt3L cultures (Fig. 1C). Notably, the expansion of cDC1 using KitL was sustained over a prolonged period without any apparent plateau until day 11 (Fig. 1C). KitL appeared to enhance cDC1 and cDC2 production to a greater extent than pDC production. Both KitL/Flt3L protocols increased cDC1 production by approximately tenfold compared to Flt3L alone on days 10 and 11 of culture.

Thrombopoietin (TPO) has been used to expand human CD34⁺ HSCs and progenitors [28, 29], and an Flt3L + TPO protocol has been shown to derive all three DC subsets from human CD34⁺ cells [30]. Therefore, we decided to test if TPO could enhance cDC1 generation from mouse BM. Compared to Flt3L alone, Flt3L + TPO culture enhanced cDC1 and pDC production by twofold and fourfold, respectively (Supporting Information Fig. 1D and F). We then attempted multiple combinations to incorporate the Flt3L + TPO protocol to our KitL/Flt3L protocol. In all combinations tested, the incorporation of TPO did not further enhance or only slightly further enhanced cDC1 or pDC production from KitL/Flt3L culture (Supporting Information Fig. 1E and G). As the major enhancement of cDC1 production came from KitL, we decided to focus on the KitL/Flt3L protocols.

KitL/Flt3L-derived cDC1 are developmentally and functionally related to in vivo cDC1

cDC1 development requires *Irf8* autoactivation mediated by the *Irf8* +32 kb enhancer, as *Irf8* +32^{-/-} mice lack cDC1 in vivo and cannot generate cDC1 in Flt3L culture [31]. We cultured BM from WT and *Irf8* +32^{-/-} mice using both the sequential and combined KitL/Flt3L protocols. WT but not *Irf8* +32^{-/-} BM generated cDC1 (Fig. 2A–C). Conversely, retroviral expression of IRF8 strongly increased cDC1/cDC2 ratios in WT BM KitL/Flt3L culture (Supporting Information Fig. 1H), consistent with previous results from Flt3L culture [18]. Therefore, cDC1 derived from KitL/Flt3L culture are developmentally similar to in vivo cDC1 in terms of IRF8 dependence.

Next, we wondered if cDC1 derived from KitL/Flt3L culture are functionally similar to in vivo cDC1 with regard to cytokine production and antigen cross-presentation. Consistent with DC subsets from spleens and Flt3L cultures, cDC1 but not cDC2 derived from KitL/Flt3L protocols could produce IL-12 in response to STAg (Fig. 3A and B, Supporting Information Fig. 2B and C). In the same experiments, we also examined DCs derived from iCD103 culture, a culture system using Flt3L plus GM-CSF that generates CD103⁺ cDC1 [32] (Supporting Information Fig. 2A). In agreement with spleens, Flt3L alone, and KitL/Flt3L cultures, iCD103-derived cDC1 but not cDC2 produced IL-12 upon STAg stimulation (Fig. 3A and B, Supporting Information Fig. 2B and C). In vitro cross-presentation assays demonstrated that only cDC1, not cDC2, derived from KitL/Flt3L cultures could induce OT-1 T-cell division upon incubation with Abelson-OVA, a form of dead cell-associated antigen (Fig. 3C–E). In summary, cDC1 generated from KitL/Flt3L cultures show functional similarities to bona fide in vivo cDC1.

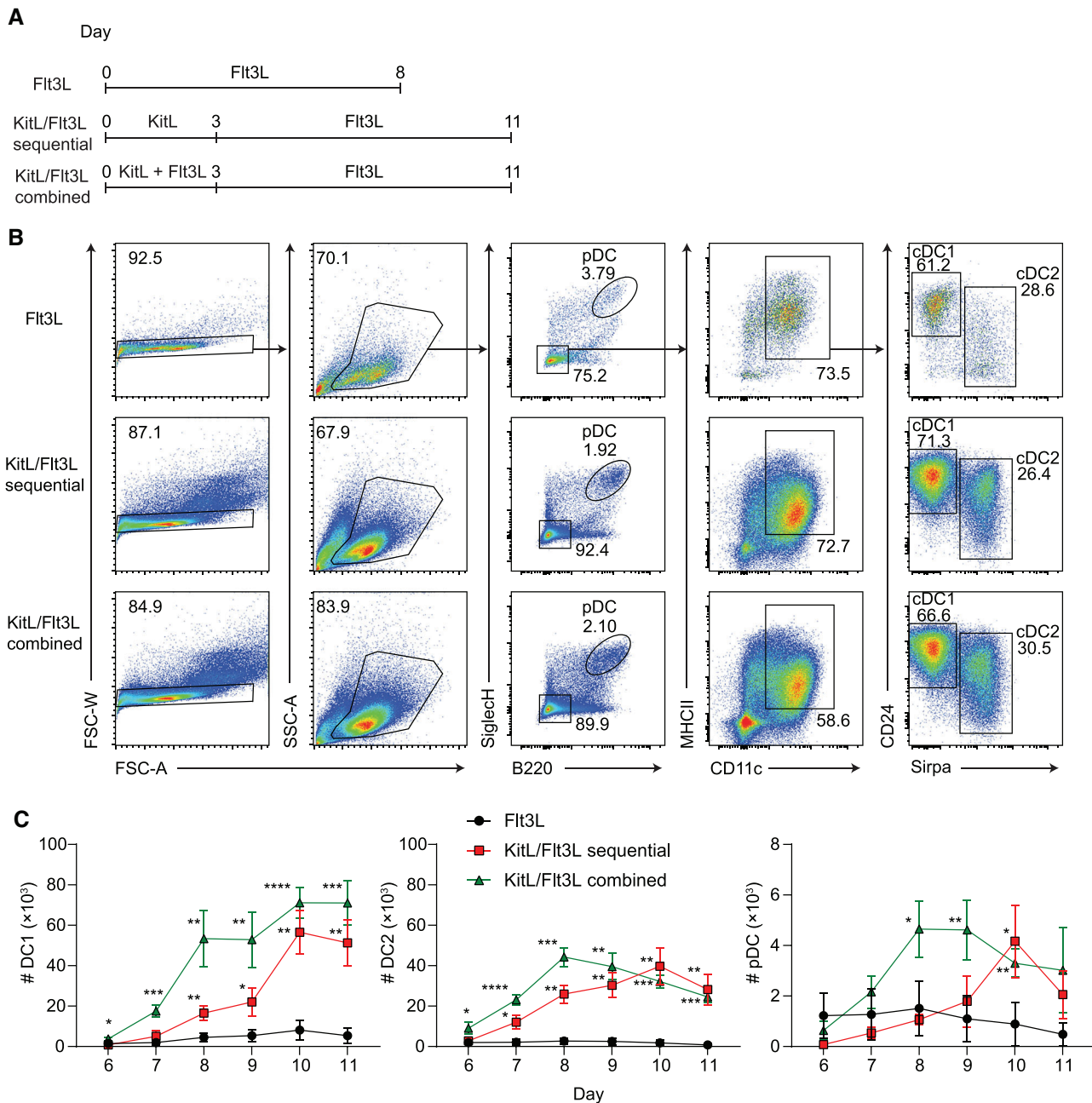


Figure 1. KitL/Flt3L cultures generate large numbers of type 1 conventional DC (cDC1), type 2 conventional DC (cDC2), and plasmacytoid DC (pDC). (A) Scheme of different BM culture methods. (B) Representative flow cytometry plots of 2.5×10^5 BM cells plated at 6.25×10^5 cells/mL and cultured with indicated methods as in (A). Data shown are one of four similar experiments. (C) Numbers of cDC1, cDC2, and pDC derived from per 10^5 BM with indicated methods at indicated time points. Data shown are one of four similar experiments with four biologically independent samples. Data are mean $\pm$ SD. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$ (one-way ANOVA).

Not all in vitro generated DCs that can cross-present in vitro are able to induce tumor regression in vivo. For example, Mo-DCs can cross-present in vitro but cannot induce tumor regression when used as an in situ vaccine in cDC1-deficient *Irf8* +32^{-/-} mice due to their inability to migrate to tDLNs [33]. In addition, although iCD103 DCs promote antitumor responses when coadministered with Flt3L, poly I:C, and irradiation [34], it was unknown whether they could induce tumor regression upon in

situ vaccination. Therefore, we tested whether KitL/Flt3L and iCD103 cDC1 could induce tumor regression in *Irf8* +32^{-/-} mice. As previously described [33], *Irf8* +32^{-/-} mice were implanted with an immunogenic fibrosarcoma, 1956-mOVA, and vaccinated intratumorally with three doses of cDC1 or phosphate buffered saline (PBS) (Fig. 4A, Supporting Information Fig. 3A). As expected, PBS-treated *Irf8* +32^{-/-} mice failed to control tumor growth and failed to prime antigen-specific CD8 T cells

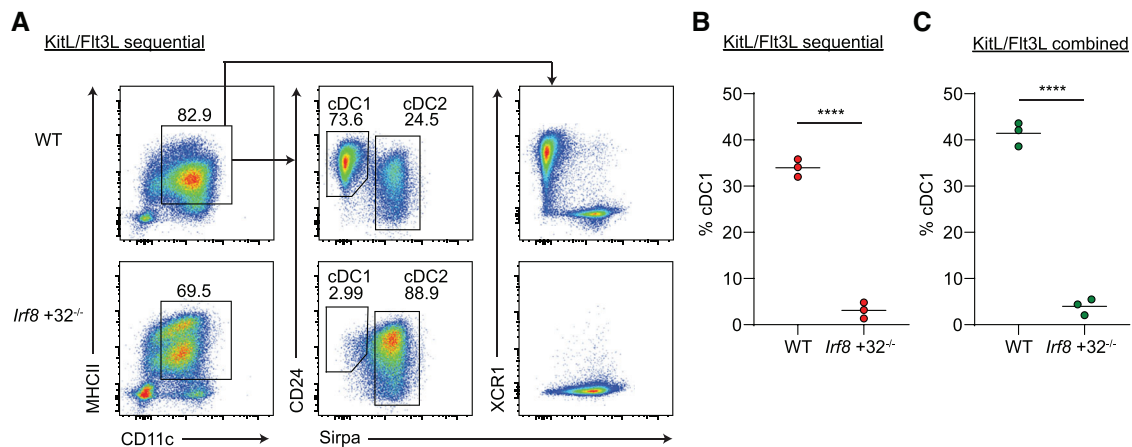


Figure 2. KitL/Flt3L type 1 conventional DC (cDC1) require the *Irf8* +32 kb enhancer. (A) Representative flow cytometry plots of sequential KitL/Flt3L cultures from WT and *Irf8* +32^{-/-} BM (pregate: B220⁻ cells). (B) % cDC1 of live singlets from WT and *Irf8* +32^{-/-} BM from indicated cultures. Data are pooled from two-independent experiments. Center values indicate mean. **p* < 0.05, ***p* < 0.01, ****p* < 0.001, *****p* < 0.0001 (unpaired, two-tailed Student's *t*-test).

(Fig. 4B–D). In contrast, antigen-specific CD8 T cells were generated in mice treated with cDC1 derived from Flt3L, KitL/Flt3L, or iCD103 cultures, and tumors regressed (Fig. 4B–D). These results suggest that, like Flt3L cDC1, KitL/Flt3L and iCD103 cDC1 can capture tumor antigens and migrate to tumor-draining lymph nodes (tdLNs) to prime CD8 T cells, resulting in tumor regression.

KitL treatment recruits Flt3⁻ HSCs and MPPs into Flt3L-mediated in vitro cDC1 generation

We hypothesized that KitL treatment allowed Flt3⁻ HSCs and progenitors to develop in vitro and acquire Flt3 expression, thus contributing to cDC1 generation. To test this, we used cell sorting to purify Flt3⁻ LT-HSC, ST-HSC, MPP2, MPP3, as well as Flt3⁺ MPP4, MDP, and CDP. We then cultured these populations with Flt3L, KitL/Flt3L, or iCD103 method (Fig. 5A–C). In Flt3L-only cultures, most Flt3⁻ cells failed to survive, and cDC1 arose only from Flt3⁺ progenitors such as MPP4, MDP, and CDP (Fig. 5E, Supporting Information Fig. 3B and C). In contrast, KitL/Flt3L culture generated cDC1 from all Flt3⁻ populations as well as Flt3⁺ MPP4 and MDP (Fig. 5D and E, Supporting Information-Fig. 3B and C). cDC1 generation from Flt3⁺ MPP4 was also significantly enhanced by KitL/Flt3L culture (Fig. 5E). iCD103 cultures showed trends toward enhanced cDC1 generation from ST-HSC and MPP4, which did not reach statistical significance. To investigate if Flt3⁻ HSCs and progenitors developed into Flt3⁺ progenitors in vitro, we performed shorter versions of Flt3L and KitL/Flt3L cultures. Consistent with observations in standard Flt3L and KitL/Flt3L cultures, short-term KitL/Flt3L treatment supported survival of Flt3⁻ HSCs and MPPs, whereas Flt3-only treatment did not (Fig. 5F). Due to exposure to high amounts of Flt3L, no surface Flt3 expression could be detected in either Flt3L or KitL/Flt3L condition (Fig. 5F and G). Upon fixing and permeabilization, Lin⁻ progenitors expressing cytoplasmic Flt3 were

identified. Notably, some originally Flt3⁻ HSCs and MPPs started to express intracellular Flt3 in KitL/Flt3L cultures (Fig. 5F and H). KitL/Flt3L culture also significantly expanded MPP4 to generate more Flt3⁺ progenitors (Fig. 5G and H). These in vitro Flt3⁺ progenitors also expressed M-CSFR, resembling in vivo MDP and CDP. In summary, these observations demonstrate that KitL can recruit Flt3⁻ HSCs and MPPs, as well as expand MPP4, thereby enhancing cDC1 generation.

Discussion

In summary, we present two KitL/Flt3L protocols that increase in vitro production of cDC1 and cDC2 by tenfold compared to the established Flt3L culture. We also showed that cDC1 generated from either KitL/Flt3L cultures or a previously published iCD103 method are sufficient for tumor regression in cDC1-deficient hosts. KitL/Flt3L protocols have the following technical distinctions from iCD103 protocols. First, KitL/Flt3L cultures generate all three DC subsets in large numbers compared to predominantly cDC1 generated by iCD103 culture [32]. Hence, the KitL/Flt3L system provides a useful tool for studying not only cDC1 but also cDC2 and pDC. Second, KitL/Flt3L protocols are somewhat shorter than iCD103 protocols, generating large numbers of cDC1 in 11 days as compared to 16 days. Moreover, our study demonstrates a clear mechanism for the enhanced cDC1 production with KitL/Flt3L protocols, where KitL supports in vitro development of Flt3⁻ HSCs and MPPs into Flt3⁺ progenitors to participate in Flt3L-mediated cDC1 generation.

Several protocols have been reported to generate human cDC1 [35–39]. These use combinations of Flt3L, KitL, GM-CSF, IL-4 (FSG4) [35, 36, 38] or combinations of cytokines and stromal cells, such as MS5 cells with Flt3L, KitL, GM-CSF (MS5 + FSG) [37, 38], and OP9-DL1 cells together with FSG [39]. Although KitL is a common component in these protocols, no mechanistic

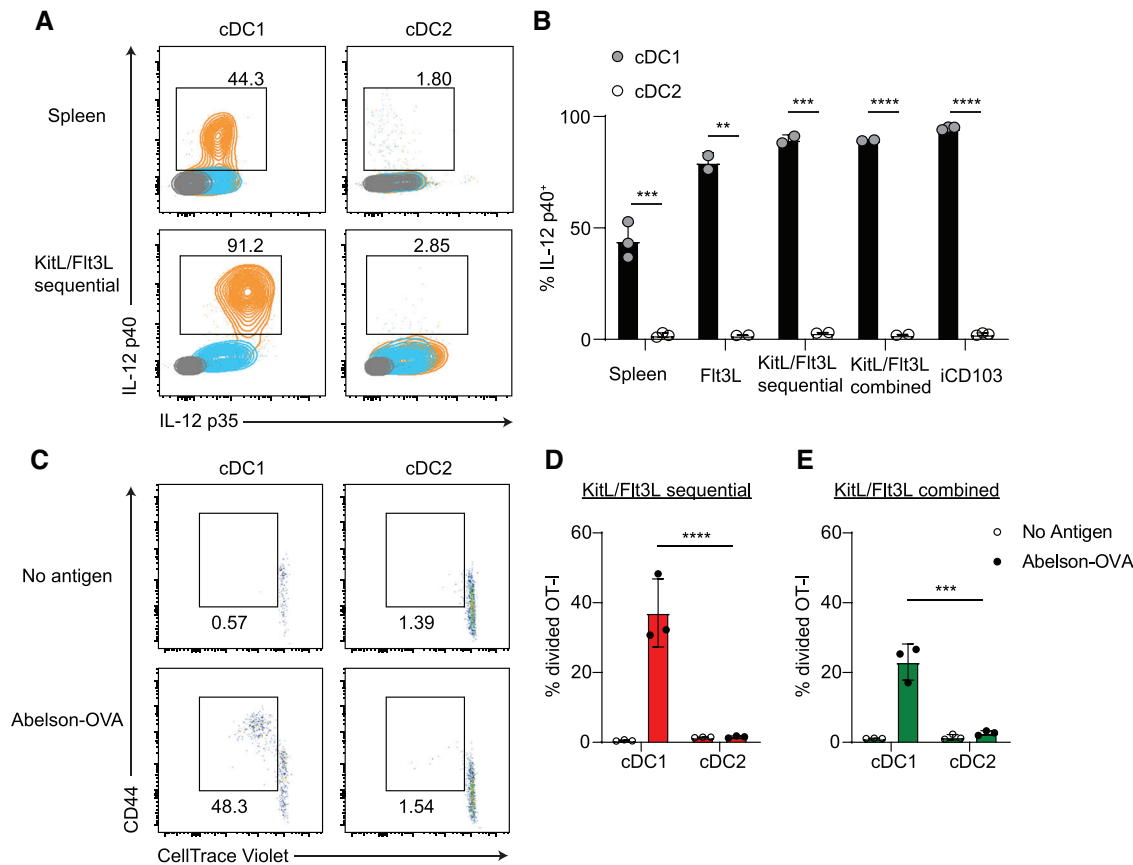


Figure 3. KitL/Flt3L type 1 conventional DC (cDC1) can produce IL-12 and can cross-present cell-associated antigen. (A) Representative flow cytometry plots of splenocytes or sequential KitL/Flt3L cultures stimulated with *Toxoplasma gondii* soluble tachyzoite antigen (STAg). Numbers represent % IL-12 p40⁺ cells within gated cDC1 and type 2 conventional DC (cDC2) populations, respectively. STAg-stimulated cells are depicted in orange, unstimulated cells are depicted in blue, and unstimulated cells stained with isotype control antibodies are depicted in gray. (B) % IL-12 p40⁺ cells within gated cDC1 and cDC2 populations from indicated cultures treated with STAg. Data are pooled from two independent experiments. (C) CellTrace Violet dilution of OT-1 T cells after in vitro exposure to indicated cDC1 or cDC2 with polyI:C alone (no antigen) or polyI:C plus Abelson-OVA (Abelson-OVA). Numbers represent % divided OT-1 T cells (pregate: CD45.2⁻ CD45.1⁺ CD8 α ⁺ V α 2⁺ cells). (D and E) % divided OT-1 T-cell from indicated conditions. Data are pooled from three independent experiments. Data are mean $\pm$ SD. * p < 0.05, ** p < 0.01, *** p < 0.001, **** p < 0.0001 (unpaired, two-tailed Student's t -test).

basis has been reported for its inclusion. Our data with mouse BM suggest that KitL may act to recruit and expand human HSCs and other Flt3⁻ progenitors into cDC1 generation. However, further studies are necessary to validate this hypothesis.

Materials and methods

Mice

WT C57BL6/J mice were obtained from the Jackson laboratory. *Irf8* +32^{-/-} mice have been described [31]. Both sexes were used between 6 and 12 weeks of age for all experiments. All in vivo experiments were performed in our specific-pathogen free facility. All experiments were performed in accordance with procedures approved by the AAALAC-accredited Animal Studies Committee of Washington University in St Louis and were in compliance with all relevant ethical regulations.

Isolation and culture of bulk BM

BM were harvested from femurs, tibias, and pelvises. Each bone was cut open on both ends and placed in a 0.5 mL microcentrifuge tube with a hole at the bottom, which is placed in a 1.5 mL microcentrifuge tube. The inner 0.5 mL tube carrying the bones is filled with 120 μ L MACS buffer (5% fetal calf serum (FCS), 2 mM EDTA in PBS), and BM were flushed into the outer 1.5 mL tube by centrifuging at 5000g for 2 min. Red blood cells were lysed using ammonium chloride-potassium bicarbonate (ACK) buffer. After RBC lysis, cells were washed with MACS buffer and then passed through a 70 μ m strainer.

For Flt3L cultures, BM were plated at 2.5×10^6 cells/mL in I10F (Iscove's Modified Dulbecco's Medium supplemented with 10% heat-inactivated FCS, L-glutamine, sodium pyruvate, MEM nonessential amino acid, penicillin/streptomycin, and 55 μ M β -mercaptoethanol) supplemented with 5% Flt3L-Fc conditioned media and cultured for 8 days unless otherwise specified.

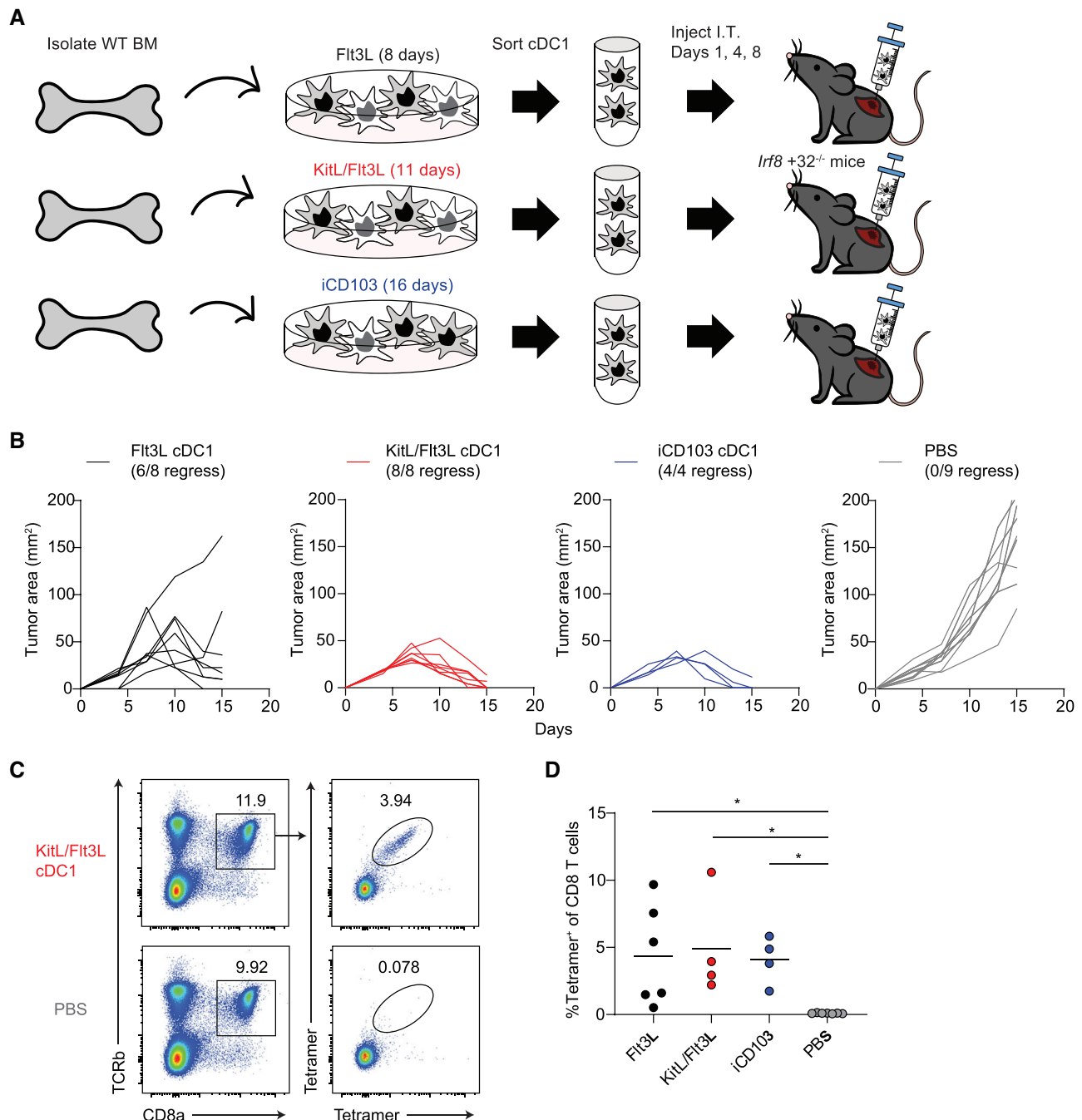


Figure 4. Both KitL/Flt3L type 1 conventional DC (cDC1) and iCD103 cDC1 can mediate tumor regression in vivo. (A) Scheme of WT BM cultures for cDC1 injections into *Lrf8* +32^{-/-} mice on day 1, 4, and 8 after tumor inoculation. I.T., intratumoral. (B) *Lrf8* +32^{-/-} mice were injected with 10⁶ 1956-mOVA fibrosarcoma and treated by three doses of 10⁶ cDC1 generated from indicated cultures or PBS. Shown are tumor growth curves for individual mice. Data are pooled from four independent experiments. (C and D) Spleens from tumor-bearing mice as in (B) were harvested on day 15 and analyzed for SIINFEKL-K^b tetramer⁺ CD8 T cells. (C) Representative flow cytometry plots from KitL/Flt3L cDC1- and PBS-treated mice, respectively (pregate: B220⁻ cells). (D) % SIINFEKL-K^b tetramer⁺ of CD8 T cells from indicated conditions. CD8 T cells are gated as B220⁻ TCRβ⁺ CD8α⁺. Data are pooled from four-independent experiments. Center values indicate mean. **p* < 0.05 (one-way ANOVA).

For sequential KitL/Flt3L cultures, BM were plated at 6.25×10^5 cells/mL in I10F supplemented with 5% KitL conditioned media and cultured for 3 days. The medium is then replaced with the same volume of I10F supplemented with 5% Flt3L-Fc conditioned media, and the cells were cultured for

another 7–8 days unless otherwise specified. Combined KitL/Flt3L cultures were performed similarly as sequential cultures except that I10F supplemented with 5% KitL conditioned media plus 5% Flt3L-Fc conditioned media were used for the first 3 days. For experiments with recombinant KitL (rKitL/Flt3L cultures),

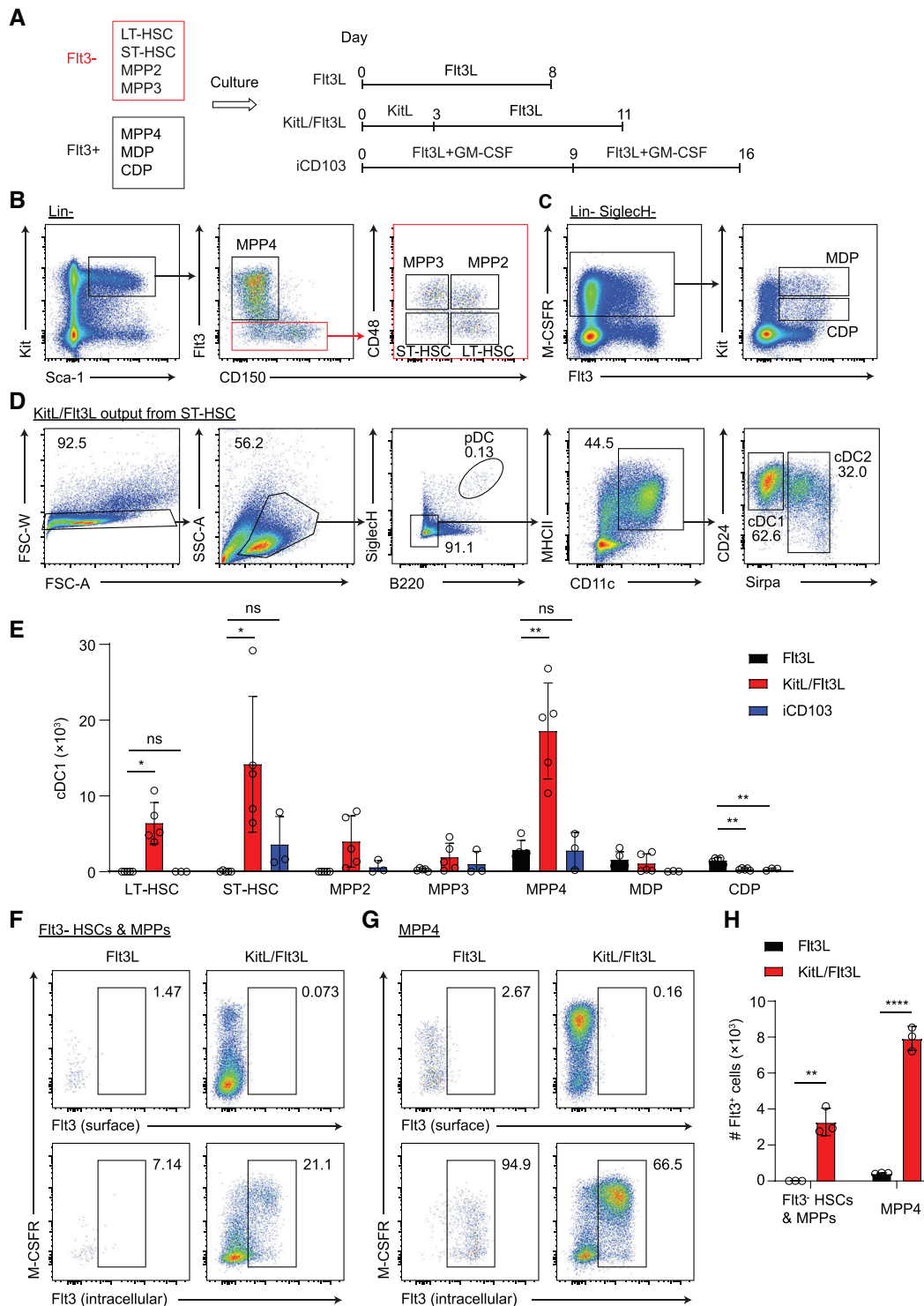


Figure 5. KitL recruit Flt3⁻ HSCs and progenitors and expand MPP4 to enhance in vitro type 1 conventional DC (cDC1) generation by Flt3L. (A) Scheme of HSC/progenitor sorting and culture. (B) Gating strategies for LT-HSC, ST-HSC, MPP2, MPP3, and MPP4. (C) Gating strategies for MDP and CDP. (D) Representative flow cytometry plots of sequential KitL/Flt3L cultures from ST-HSC. (E) Numbers of cDC1 generated from per 10³ indicated HSCs or progenitors under culture conditions depicted in (A). Data are pooled from four independent experiments. (F-H) Flt3⁻ HSCs and multipotent progenitors (MPPs) (indicated by the red gate in (B)) or MPP4 were sorted and cultured in Flt3L for 2 days (Flt3L) or in KitL for 3 days and then in Flt3L for 2 days (KitL/Flt3L) and analyzed for surface or intracellular Flt3 expression. (F and G) Representative flow cytometry plots showing surface (top) and intracellular (bottom) Flt3 expression (pregate: Lin⁻ cells). (H) Numbers of intracellular-Flt3⁺ cells generated from per 10³ indicated populations under indicated conditions. Data are mean ± SD. *p < 0.05, **p < 0.01, ***p < 0.001, ****p < 0.0001 (one-way ANOVA in (E), unpaired, two-tailed Student's t-test in (H)).

BM were plated in I10F supplemented with 200 ng/mL recombinant KitL (PeproTech) and cultured for 3 days. The medium is then replaced with the same volume of I10F supplemented with 5% Flt3L-Fc conditioned media, and the cells were cultured for another 8 days.

iCD103 cultures were performed as described previously [32] but with a few modifications. Briefly, BM were plated at 7.5×10^5 cells/mL in I10F supplemented with 5% Flt3L-Fc conditioned media and 2.5 ng/mL GM-CSF (PeproTech) and cultured for 9 days. Nonadherent cells were then replated at 1.5×10^5 cells/mL in I10F supplemented with 5% Flt3L-Fc conditioned media plus 2.5 ng/mL GM-CSF for another 7 days unless otherwise specified.

For conditions with TPO, 50 ng/mL recombinant murine TPOs (PeproTech) were used.

Flow cytometry and antibodies

Flow cytometry was performed using a FACSARIA Fusion (BD Biosciences). Data were analyzed using FlowJo software. Surface staining was performed at 4°C in the presence of Fc block (2.4G2, BD Biosciences) in MACS buffer.

For the analysis of IL-12 p35 and p40, splenocytes or cultured cells were plated at 5×10^5 cells/mL and stimulated with 1 µg/mL STAg for 6 h in the presence of 250 ng/mL Brefeldin A. The cells were stained for surface markers, fixed with 2% paraformaldehyde in PBS, washed with PBS, and then permeabilized with 0.5% saponin while staining for IL-12 p35 and p40 at 4°C overnight. The procedure for Flt3 intracellular staining was similar to IL-12 p35, p40 intracellular staining except that in the last step, cells were permeabilized with 0.5% saponin while staining for Flt3 at room temperature for 1 h.

The following antibodies/stains were used from BD Biosciences: 7AAD, Kit (2B8), Flt3 (A2F10.1), Clec9A (10B4), streptavidin-FITC; Biolegend: CD3ε (145-2C11), CD11b (M1/70), CD11c (N418), CD19 (6D5), B220 (RA3-6B2), IL-7Rα (A7R34), Ly6C (HK1.4), Ly6G (1A8), Ter119 (Ter-119), MHCII (M5/114.15.2), CD8α (53-6.7), CD24 (M1/69), CD48 (HM48-1), M-CSFR (AFS98), CD150 (TC15-12F12.2), Sirpα (P84), XCR1 (ZET), Sca-1 (E13-161.7), SiglecH (551), TCRβ (H57-597), CD103 (2E7), streptavidin-PE, CD45.1 (A20), CD45.2 (104), Vα2 (B20.1); eBioscience: IL-12/IL-23 p40 (C17.8), rat IgG2a isotype control (sBR2a), streptavidin-APC; Invitrogen: CD11c (N418) CD105 (MJ7/18); Tonbo Biosciences: MHCII (M5/114.15.2); R & D systems: IL-12/IL-35 p35 (27537), mouse IgG1 isotype control (11711).

Isolation and culture of BM HSCs and progenitors

BM were isolated as described above and depleted Lin⁺ cells by staining with corresponding biotinylated antibodies followed by depletion with MojoSort Streptavidin Nanobeads (Biolegend). The remaining cells were stained with fluorescent antibodies/stains before sorting. Lin markers for ST-HSC, LT-HSC, MPP2,

and MPP3 sorting include CD3ε, CD11b, CD11c, CD19, B220, IL-7Rα, Ly6C, Ly6G, Ter119, and MHCII. Lin markers for MPP4, MDP and CDP sorting include CD3ε, CD11c, CD19, B220, CD105, IL-7Rα, Ly6G, Ter119, and MHCII. A FACSARIA Fusion was used for sorting, and cells were sorted into I10F and kept at 4°C until plating. A number of $1.5\text{--}2.5 \times 10^3$ cells were plated in 200 µL in U-bottom 96-well plates and cultured as described in the isolation and culture of bulk BM section. For experiments that investigate Flt3⁺ progenitor generation, shorter versions of the culture protocols were used. In those experiments, Flt3L cultures were done for 2 days, and KitL/Flt3L cultures involved 3 days of KitL treatment followed by 2 days of Flt3L treatment.

Tumor lines and growth experiments

Tumor lines were used, and growth experiments were performed as described [33]. Briefly, 1956-mOVA fibrosarcomas derived from frozen stocks were propagated for 2 days in RPMI media supplemented with 10% FCS, washed three times with PBS, and resuspended at 1×10^7 cells/mL in PBS. *Irf8*^{+32^{-/-}} mice were subcutaneously injected into the flanks with 10^6 tumor cells. Tumor growth was measured with a caliper, and tumor area was calculated by the multiplication of two perpendicular diameters. In accordance with our IACUC-approved protocol, maximal tumor diameter was 20 mm in one direction and in no experiments was this limit exceeded. For intratumoral DC injections, 1×10^6 sorted cDC1 were injected intratumorally on days 1, 4, and 8 after tumor implantation, and tumor growth was measured as described above. For analysis of OVA-specific CD8 T cells, spleens were harvested 15 days after tumor transplantation, and SIINFEKL-H2-K^b staining was performed as described [33].

Retroviral transduction

Retroviral constructs expressing Thy1.1 and IRF8-T2A-Thy1.1 have been described [18]. For retroviral packaging, Plat-E cells were plated in 6-well plates at 0.4×10^6 cells/mL and incubated overnight. Retroviral plasmids mixed with TransIT-LT1 (Mirus Bio) in Opti-MEM reduced serum medium were transfected into the Plat-E cells. The culture media were changed the next day, and the supernatant containing retroviruses were collected 48 h posttransfection. For retroviral transduction, ACK-lysed BM were cultured in I10F supplemented with 5% KitL conditioned media or 5% KitL conditioned media together with 5% Flt3L-Fc conditioned media overnight. After removing the culture media, the cells were transduced with the supernatant containing retroviruses in the presence of 2 mg/mL polybrene by centrifugation at $729 \times g$ for 1 h at room temperature. The culture media were changed 24 h later, and the cells were further cultured following the KitL/Flt3L protocols as described in the isolation and culture of bulk BM section.

In vitro cross-presentation assays

For the preparation of Abelson-OVA, MHCI TKO BM transformed into Abelson TKO cell line with v-abl supernatant. Abelson TKO cell line was transduced with MSCV-mOVA-IRES-Thy1.1 retrovirus vector and sorted for Thy1.1⁺ cells. The cells were cultured and then killed by 3 cycles of freezing in liquid nitrogen and thawing at 37°C.

For in vitro cross-presentation assays, sorted OT-1 T cells were labeled with 1 µM CellTrace Violet (Invitrogen). A number of 12,500 labeled OT1 T cells were then cocultured with 12,500 sorted cDC1 or cDC2 at the presence of 125 ng/mL polyI:C (Sigma-Aldrich) with or without 25,000 Abelson-OVA in U-bottom 96-well plates. Cultures were analyzed on day 3 for CellTrace Violet dilution.

Statistics

Statistical analysis was performed using GraphPad Prism software version 8. When two groups were compared, unpaired, two-tailed Student's *t*-test was used to determine significant differences. When multiple groups were compared, one-way ANOVA was used to determine significant differences followed by post hoc comparisons. Means ± SD are displayed in scatter plots with bars, and means are displayed in scatter plots without bars.

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Abbreviations: cDC1: type 1 conventional DC · cDC2: type 2 conventional DC · Flt3: Fms-like tyrosine kinase 3 · Flt3L: Flt3 ligand, KitL: Kit ligand · LT-HSC: long-term HSC · MPP: multipotent progenitor · pDC: plasmacytoid DC · STAg: *Toxoplasma gondii* soluble tachyzoite antigen · ST-HSC: short-term HSC · tdLN: tumor-draining lymph node · TPO: thrombopoietin

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